



Cell: +1 (201) 707 5797

Email:

mainak.office13@gmail.com

LinkedIn:

<https://www.linkedin.com/in/mainakdastech>

Location:

New jersey, USA

Most recent Roles

- ✓ Technical Lead at Goldman Sachs
- ✓ Technical Architect in UBS

Education

– Bachelor of Engineering,
Jadavpur University,
India

Mainak Das

Technical Architect | API Development

PROFESSIONAL SUMMARY

Skilled Technical Lead / Architect with 20 years of experience in the design, development, and implementation of distributed microservice based applications specializing in Asset and Wealth Management Systems. Extensive expertise in Collateral Management (Apex Collateral from FIS), Corporate Actions (XSP from FIS), and Portfolio Accounting (Invest One) System as well as different products ranging from Fixed Income to Equities. Proven track record in modernizing legacy systems, leveraging AWS cloud technologies, and driving operational efficiency.

Experienced in design and development with strong skills in Java, Spring Boot, and REST API development. Proficient in AWS cloud services (S3, Lambda, EC2, EBS, DMS, API Gateway) and modern messaging like Kafka, and relational databases (Sybase, Oracle). Followed organizational guidance in designing new applications.

Certified AWS Architect and CFA Investment Foundations, with a deep understanding of financial services and the ability to innovate and optimize business processes. Demonstrated leadership in managing global teams and collaborating across business units, ensuring successful project outcomes. Adept in Agile/Scrum methodologies to deliver projects on time and align with strategic business goals. Focused on supporting the ongoing evolution of financial systems and enhancing business functionality through technology-driven solutions.

EXPERTISE / CERTIFICATIONS

- Technical Expertise

- Hands-on experience with Java, Python, Sybase, Snowflake, Oracle, Spring Boot REST Services, Kafka, XML, XSD, Autosys, AWS, Open API, Swagger
- Programming Languages - Java, SQL, Python
- Cloud Platforms - AWS (S3, Lambda, EC2, EBS, Aurora, API Gateway, DMS)
- IAC Platforms – Typescript, aws-cdk
- Service Observability - Prometheus (On -Premises)
- Big Data Frameworks – Kafka, Apache Spark, Spring Boot
- Databases: MySQL, Sybase, Oracle,
- Data Visualization – Qlik, Alteryx
- Project Management Tools – Agile / Scrum, JIRA

- Certifications

- AWS Certified Solutions Architect - Associate
- CFA Investment Foundations Certified
- AWS Certified Cloud Practitioner
- IBM SOA (Service Oriented Architecture) Certified

- Functional Expertise

- Extensive experience working in Investment banking environment across Front, Middle and Back Offices.
- Detail requirement understanding with Desks for business requirements on different products and processes.
- Experienced working with following products
 - Treasuries,
 - Equities,
 - Corporate Bonds,
 - Options
 - MBS

- Global Team Management:

- Skilled in managing teams across multiple continents, fostering collaboration and productivity in diverse, multicultural environments.
- **Business Coordination and Project Prioritization**
 - Adept at understanding critical business requirements and aligning team efforts to meet strategic objectives
 - Expertise in prioritizing projects based on resource constraints and criticality, ensuring optimal use of resources
- **Banking Software Proficiency**
 - Experienced in Finacle Banking, a leading banking software solution
- **Process Optimization**
 - Skilled in optimizing processes to enhance efficiency and outcome

DOMAIN EXPERTISE & EXPOSURE

- Investment Banking
- Asset and Wealth Management
- API Development
- Shared Services
- Time & Expense Management
- Stakeholder and Delivery Management with onsite-offshore co-ordination

PROFESSIONAL EXPERIENCE WITH GOLDMAN SACHS SERVICES LLC

OCT 2015 – CURRENT

Role: Project / Product Lead

Significant Engagement Description:

- Tech Lead in Portfolio Management Database (PMDB) Team

Implementing critical business requirements like providing efficient “Start of the Day” Positions / Transactions / Tax Lots data to other systems like Trading Desk, Guidelines, Compliance, Reconciliation, Real Time Position Service Systems within GSAM division. This data is held as part of Shadow Booking against Custodians Trades. Implemented very high performing data load and enrichment process leveraging Service APIs on Spring Rest Template, Java Multi Threading, Asynchronous messaging framework including Kafka and efficient DB operations.

PMDB stores accounting portfolio data primarily sourced from the Invest One accounting system. Including positions, transactions, products, and open/closed tax lots. It is replicated through native Sybase replication to downstream DBs across Asset Management divisions. The data is crucial as it provides the "Start of the Day" view for trading desks and is also used by Compliance and Guidelines teams for ongoing business processes throughout the day. This
- Development and Managing Product API Data

Involved in Design and development of Product data service through Service API which converts Financial Instruments data from middle office systems into different formats accepted by downstream. The Product scope includes a vast range of asset groups including Treasury (Bills, Notes, Bonds) , Corporate Bonds, Equity, Mortgage based Securities (MBS), and Options (Call / Put).. The Service implementation involves both Synchronous and Asynchronous implementation depending on the users' request. It involves Kafka, SpringBoot, Open API, Swagger, Business logic implementation with Java and Python.
- Involved in Designing Application Modernization by leveraging AWS Cloud

Involved in designing scalable, reliable and durable systems by modernizing the legacy PMDB application by migrating it to the AWS Cloud leveraging advanced cloud technologies like Data Management Services (DMS), AWS Aurora, Lambda, API Gateway, and EC2. The goal is to load IBOR data into the AWS Aurora database through Java/python-based logic deployed over EC2 / Lambda and scaled through ASG and load balancers. Also designed data migration from Aurora database to/from on-prem Sybase database through AWS data migration service a.k.a. DMS for both bulk load as well as CDC (Change Data Capture).

- Design and developed Data Segregation across GSAM data

This project focuses on segregating portfolio accounting data between Public and Private Accounts within GSAM as part of strict compliance guidance.

Involved in designing and implementing System segregation across different PMDB and downstream applications which enforces strict segregation rules to differentiate users on the Public and Private sides of the business. Extensively involved in active requirement discussions with different stakeholders including Desks, Compliance, Risk and various downstream system and designing the data model for Private side of the business from PMDB data perspective (Position / Transaction / Tax Lots / NAV/ Product) and implementing the business logic through Java, SQL, File based processing to segregate data at account level..

- Implementing and optimizing ETL environment for Collateral Management and Corporate Actions System

This project involves installing a vendor-provided Collateral Management tool and integrating it with data from external systems, including Fixed Income Securities, Equity trades, and other static data.

Involved in design, implementation and optimization of various Java based extract, transform and load processes to feed data into Collateral Management as well as Corporate Actions Systems from multiple In-house data stores like Account, Product, Trades, Positions. Implemented very high performant multi-threaded system using Thread Latch / Semaphore in Java to extract and format the data within SLO.

Responsibilities:

- Implemented microservices based Services leveraging both Synchronous and Asynchronous communication implemented by Spring Boot, Kafka, Java
- Optimized performance of an existing ETL process for Corporate Actions System by implementing Java multi-threaded environment.
- Extensive experience in using different build tools like Maven, Gradle and deploying the code through firm approved CI/CD framework
- Implemented high performant REST Based Service APIs to provide data for different Fixed Income (Treasuries, Corporate Bonds) as well as Equity products for downstream Systems to consume. Implementation was based on Spring Boot, Open API and Java based business logic.
- Designed and implemented modernization of Legacy process to AWS Cloud leveraging EC2, EBS, S3, Lambda, Event Bridge, SNS, DynamoDB, AWS Aurora and DMS.
- Designed and implemented Java Multi-threaded environment including Executor thread pool / Latch / Semaphore for a high performant data extraction and formatter tool which feed data to Collateral management system on daily basis.
- Extensively used Java 8 features such as lambda expressions, Parallel operations on collections, multithreading
- Extensive experience in Agile working culture.
- and for effective sorting mechanisms and Streams to store and process the data.
- Hands on experience on different scheduling tools like Autosys, Spring Scheduler.
- Coordinated with business regularly to align technology work with Business Requirement.

Technologies / Environment

- Java, Python, JSON, AVRO
- Spring, SpringBoot, REST Services, Microservices.
- Apache Kafka, JMS
- Maven, Gradle,
- AWS Compute including EC2, Lambda, AWS Storage like S3. EBS, AWS Databases like RDS AWS Aurora, AWS Data Migration System like DMS (both bulk load and Change Data Capture or CDC),
- Sybase, Oracle, MYSQL
- Tomcat

PROFESSIONAL EXPERIENCE WITH **WIPRO TECHNOLOGIES**

NOV 2010 - OCT 2015

Client: UBS Bank

Role: Technical Architect / Project Lead

Significant Engagement Description:

- Application stack modernization through automation

This project is about migration of business applications from WAS 6.1 to WAS 8.5 or Tomcat and automating the migration process to derive better ROI. These applications involved various functionalities including Position reconciliation, Fraud detection, Notification etc.

- Involved in implementing an automation tool in JAVA with Eclipse native APIs, XML, XSD, JSON. This tool automates different aspects of migration thus reducing overall migration time by 40%.
- Extensively worked on day-to-day deliverables in transferring the applications code compatible to target environments.
- Standardizes overall build process in new infrastructure through CI/CD build including Maven and integrating the build process with Sonar to detect code vulnerabilities.
- Worked on migrating applications from Struts 1.2 to 2.3

Client: Bank of New York Mellon, New York

Role: Technical Lead

Significant Engagement Description:

- Report migration from Crystal Framework to Jasper

This report transformation project focuses on migrating Client's Reporting platform from Crystal to Jasper within a very short span of time to meet broader project timelines. These reports involve Stock holdings per client, Custodian Reports etc.

- Involved in design and implement a Java based migration tool which reads Crystal Reports and converts each components to Jasper format so that it reduces lot of manual works thus delivering the project at record time. Once implemented in production this tool reduced the Reports migration timeline by 70%.

Client: State Street, Boston

Role: Technical Lead

Significant Engagement Description:

- Application modernization leveraging Cloud

As part of the application modernization initiative, this project was about migrating State Street's applications running in bare metal box to private cloud platform.

- Involved in discussing application functionalities with app owners and preparing design plans for efficient migration.
- Involved in migration activities for both design and implementation for applications leveraging various technologies like Java, XML, XSD, Spring, Struts MVC, Oracle
- Also implemented Service layer in front of Database to replace all direct DB Connections through REST API. This involved application integration with various Databases seamless in target cloud environment

Responsibilities:

- Designing and Developing the Host Transformation Solution tool and Crystal to Jasper Report transformation tool to expedite the overall migration tasks which helped in winning this project.
- Extensive hands on in fixing any migration issues.
- Scratch development of web services while deploying in Tomcat environment.
- Designing the Maven build structure to standardize the build process.
- Adhering to SDLC processes in the overall migration.

- Designing externalization process so that same code can be deployed seamlessly in different environments.
- Communicating with client to understand any additional requirements
- Estimation of the project size
- Estimating the complexity of applications
- Leading / mentoring a team to do a POC which helped to win the big project.

Technologies:

- Java
- IBM WebSphere, Tomcat
- SOAP / REST
- Jasper, Crystal
- IBM MQ
- Spring, SpringBoot, Struts
- Autosys

PROFESSIONAL EXPERIENCE WITH **PRICEWATERHOUSECOOPERS PVT LTD**

FEB 2006 - OCT 2010

Client: PricewaterhouseCoopers, USA

Role: Technical Lead / Technical Developer

Significant Engagement Description:

- Shared Services Application development

This project involved design, implementation and maintenance of a bucket of applications ranging from UI applications to WebLogic Workshop based services. Primary technologies involved were WebLogic IDE, XML, XSD, Oracle, SOAP Web service

Responsibilities:

- Worked as the onsite coordinator in Tampa, Florida, USA for the team for around one year. As the onsite coordinator he was responsible for meeting with the business, gathering requirements, transferring the requirements to offshore and providing support in case of no-wait time was available. Being the offshore face his responsibility included maintaining a cordial relationship with the onshore business.
- Leading an offshore team with high performance output.
- Develop new applications and maintain the existing ones.
- Key member of the team responsible for designing new applications and exploring new technologies specific to Web Service Security.

PROFESSIONAL EXPERIENCE WITH **INFOSYS TECHNOLOGIES**

SEP 2004 - FEB 2006

Client: ANZ Bank Melbourne, Australia

Role: Technical Developer

Significant Engagement Description:

- Implementing Finacle Internet Banking for ANZ Bank

Finacle, the Universal Banking Solution from Infosys Technologies Limited has been customized as part of the requirement from ANZ Bank.

Responsibilities:

- Developed different Internet Banking modules like Loan application processing, overdraft calculation etc.
 - Requirement understanding from Tech Leads and implementing the same with high quality
 - Writing extensive unit test cases
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